Algebra - 0 2026 Spring

Homework 4

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*All of my homework is available at `https://github.com/lacampanella233/Homework.git`.

$\boxed{\textbf{Problem 1} \text{ (3E-1).}}$ Suppose T is a function from V to W. The graph of T is the subset of $V \times W$ defined by

$$\text{graph of } T = \{(v, Tv) \in V \times W : v \in V\}.$$

Prove that T is a linear map if and only if the graph of T is a subspace of $V \times W$.

Proof. Denote the graph of T by $G \subset V \times W$. On the one hand, if T is a linear map, then for and $(v_1, Tv_1) \in G$ and $(v_2, Tv_2) \in G$, and $\lambda \in \mathbb{F}$, we have

$$(v_1, Tv_1) + (v_2, Tv_2) = (v_1 + v_2, T(v_1 + v_2)) \in G,\ \lambda(v_1, Tv_1) = (\lambda v_1, T(\lambda v_1)) \in G.$$

Hence, G is a subspace of $V \times W$.

On the other hand, if G is a subspace of $V \times W$, then for any $v_1, v_2 \in V$ and $\lambda \in \mathbb{F}$, we have

$$(v_1, Tv_1) + (v_2, Tv_2) = (v_1 + v_2, Tv_1 + Tv_2) \in G,\ \lambda(v_1, Tv_1) = (\lambda v_1, \lambda Tv_1) \in G,$$

which implies that $T(v_1 + v_2) = Tv_1 + Tv_2$ and $T(\lambda v_1) = \lambda Tv_1$. Hence, T is a linear map. □

$\boxed{\textbf{Problem 2} \text{ (3E-9).}}$ Prove that a nonempty subset A of V is a translate of some subspace of V if and only if

$$\lambda v + (1 - \lambda)w \in A$$

for all $v, w \in A$ and all $\lambda \in \mathbf{F}$.

Proof. On the one hand, if A is a translate of U , then there exists $t \in V$ such that $A = t + U$. For any $v, w \in A$ and $\lambda \in \mathbf{F}$, we have $v = t + u_1$ and $w = t + u_2$ for some $u_1, u_2 \in U$. Then

$$\lambda v + (1 - \lambda)w = \lambda(t + u_1) + (1 - \lambda)(t + u_2) = t + (\lambda u_1 + (1 - \lambda)u_2) \in t + U.$$

On the other hand, if $\lambda v + (1 - \lambda)w \in A$ for all $v, w \in A$ and all $\lambda \in \mathbb{F}$, We will show that there exists $t \in V$ and $U \subset V$ such that $A = t + U$.

Pick $a_0 \in A$ and $a_0 = t + u_0$. Denote

$$U = \{a - a_0 : a \in A\}.$$

It suffices to prove U is a subspace of V .

(a) U is closed under addition. That's because for u_1, u_2 in U , set $a_1 = a_0 + u_1, a_2 = a_0 + u_2$. Then $a_1/2 + a_2/2 = a_0 + (u_1 + u_2)/2 \in A$, hence $2(a_1/2 + a_2/2) + (1-2)a_0 = a_0 + (u_1 + u_2) \in A$, which implies $u_1 + u_2 \in U$.

(b) U is closed under scalar multiplication. That's because for $u \in U$ and $\lambda \in \mathbb{F}$, set $a = a_0 + u$. Then $\lambda a + (1-\lambda)a_0 = a_0 + \lambda u \in A$, which implies $\lambda u \in U$.

□

$\boxed{\textbf{Problem 3}\text{ (3E-14).}}$ Suppose U and W are subspaces of V and $V = U \oplus W$. Suppose $w_1, \cdots, w_m$ is a basis of W. Prove that $w_1 + U, \cdots, w_m + U$ is a basis of V/U.

Proof. First we will show that $w_1 + U, \cdots, w_m + U$ is linearly independent. Suppose $\lambda_1(w_1 + U) + \cdots + \lambda_m(w_m + U) = 0 + U$ for some $\lambda_1, \cdots, \lambda_m \in \mathbb{F}$. Then $\lambda_1 w_1 + \cdots + \lambda_m w_m \in U$. Since $U \cap W = \{0\}$, we have $\lambda_1 w_1 + \cdots + \lambda_m w_m = 0$, which implies $\lambda_1 = \cdots = \lambda_m = 0$. Hence, $w_1 + U, \cdots, w_m + U$ is linearly independent.

Next we will show that $w_1 + U, \cdots, w_m + U$ spans V/U . For any $v + U \in V/U$, we can write $v = u + w$ for some $u \in U$ and $w \in W$. Since $w_1, \cdots, w_m$ is a basis of W , we can write $w = \lambda_1 w_1 + \cdots + \lambda_m w_m$ for some $\lambda_1, \cdots, \lambda_m \in \mathbb{F}$. Then

$$v + U = u + w + U = \lambda_1(w_1 + U) + \cdots + \lambda_m(w_m + U).$$

Hence, $w_1 + U, \cdots, w_m + U$ spans V/U .

Combining the above two paragraphs, we conclude that $w_1 + U, \cdots, w_m + U$ is a basis of V/U . □

$\boxed{\textbf{Problem 4}\text{ (3E-15).}}$ Suppose U is a subspace of V and $v_1 + U, \cdots, v_m + U$ is a basis of V/U and $u_1, \cdots, u_n$ is a basis of U. Prove that $v_1, \cdots, v_m, u_1, \cdots, u_n$ is a basis of V.

Proof. First we will show that $v_1, \cdots, v_m, u_1, \cdots, u_n$ is linearly independent. Suppose $\lambda_1 v_1 + \cdots + \lambda_m v_m + \mu_1 u_1 + \cdots + \mu_n u_n = 0$ for some $\lambda_1, \cdots, \lambda_m, \mu_1, \cdots, \mu_n \in \mathbb{F}$. Then $\lambda_1(v_1 + U) + \cdots + \lambda_m(v_m + U) = 0 + U$, which implies $\lambda_1 = \cdots = \lambda_m = 0$. Hence, $\mu_1 u_1 + \cdots + \mu_n u_n = 0$, which implies $\mu_1 = \cdots = \mu_n = 0$. Therefore, $v_1, \cdots, v_m, u_1, \cdots, u_n$ is linearly independent.

Next we will show that $v_1, \cdots, v_m, u_1, \cdots, u_n$ spans V . Since $v_1 + U, \cdots, v_m + U$ is a basis of V/U , we can write $v + U = \lambda_1(v_1 + U) + \cdots + \lambda_m(v_m + U)$ for some $\lambda_1, \cdots, \lambda_m \in \mathbb{F}$. Then

$$v = \lambda_1 v_1 + \cdots + \lambda_m v_m + u$$

for some $u \in U$. Since $u_1, \cdots, u_n$ is a basis of U , we can write $u = \mu_1 u_1 + \cdots + \mu_n u_n$ for some $\mu_1, \cdots, \mu_n \in \mathbb{F}$. Then

$$v = \lambda_1 v_1 + \cdots + \lambda_m v_m + \mu_1 u_1 + \cdots + \mu_n u_n.$$

Hence, $v_1, \cdots, v_m, u_1, \cdots, u_n$ spans V .

Combining the above two paragraphs, we conclude that $v_1, \cdots, v_m, u_1, \cdots, u_n$ is a basis of V . $\square$

$\boxed{\textbf{Problem 5}\text{ (3E-18).}}$ Suppose that U is a subspace of V such that V/U is finite-dimensional.

(a) Show that if W is a finite-dimensional subspace of V and $V = U + W$, then $\dim W \geq \dim V/U$.

(b) Prove that there exists a finite-dimensional subspace W of V such that $\dim W = \dim V/U$ and $V = U \oplus W$.

Proof. (a) Suppose W is a finite-dimensional subspace of V and $V = U + W$. Then the map $T : W \to V/U$ defined by $Tw = w + U$ is a surjective linear map. Hence, we have

$$\dim W = \dim \ker T + \dim V/U \geq \dim V/U.$$

(b) Since V/U is finite-dimensional, we can pick a basis $v_1 + U, \cdots, v_m + U$ of V/U . Denote $W = \operatorname{span}\{v_1, \cdots, v_m\}$. We will show that $V = U \oplus W$ and $\dim W = \dim V/U$.

First we will show that $V = U + W$. For any $v \in V$, we can write $v + U = \lambda_1(v_1 + U) + \cdots + \lambda_m(v_m + U)$ for some $\lambda_1, \cdots, \lambda_m \in \mathbb{F}$. Then

$$v = \lambda_1 v_1 + \cdots + \lambda_m v_m + u$$

for some $u \in U$. Hence, $V = U + W$.

Next we will show that $U \cap W = \{0\}$. Suppose $u \in U \cap W$. Then $u = \lambda_1 v_1 + \cdots + \lambda_m v_m$ for some $\lambda_1, \cdots, \lambda_m \in \mathbb{F}$. Hence, we have

$$\lambda_1(v_1 + U) + \cdots + \lambda_m(v_m + U) = 0 + U,$$

which implies $\lambda_1 = \cdots = \lambda_m = 0$. Therefore, $u = 0$ and $U \cap W = \{0\}$.

Combining the above two paragraphs, we conclude that $V = U \oplus W$.

Finally, since $v_1 + U, \cdots, v_m + U$ is a basis of V/U , we have $\dim W = \dim \operatorname{span}\{v_1, \cdots, v_m\} = m = \dim V/U$. $\square$

$\boxed{\textbf{Problem 6}\text{ (3F-5).}}$ Suppose $T \in \mathcal{L}(V, W)$ and $w_1, \cdots, w_m$ is a basis of $\operatorname{im} T$. Hence for each $v \in V$, there exist unique numbers $\varphi_1(v), \cdots, \varphi_m(v)$ such that

$$Tv = \varphi_1(v)w_1 + \cdots + \varphi_m(v)w_m,$$

thus defining functions $\varphi_1, \cdots, \varphi_m$ from V to $\mathbb{F}$. Show that each of the functions $\varphi_1, \cdots, \varphi_m$ is a linear functional on V.

Proof. Without loss of generality, we only need to show that φ_1 is a linear functional on V . For any $v_1, v_2 \in V$ and $\lambda \in \mathbb{F}$, we have

$$\varphi_1(v_1 + v_2)w_1 + \cdots = v_1 + v_2 = (\varphi_1(v_1) + \varphi_1(v_2))w_1 + \cdots \implies \varphi_1(v_1 + v_2) = \varphi_1(v_1) + \varphi_1(v_2),$$

$$\lambda\varphi_1(v_1)w_1 + \cdots = \lambda v_1 = \varphi_1(\lambda v_1)w_1 + \cdots \implies \varphi_1(\lambda v_1) = \lambda\varphi_1(v_1).$$

$\square$

Problem 7 (3F-8). Suppose $v_1, \cdots, v_n$ is a basis of V and $\varphi_1, \cdots, \varphi_n$ is the dual basis of V'. Define $\Gamma : V \to \mathbb{F}^n$ and $\Lambda : \mathbb{F}^n \to V$ by

$$\Gamma(v) = (\varphi_1(v), \cdots, \varphi_n(v))$$

and

$$\Lambda(a_1, \cdots, a_n) = a_1 v_1 + \cdots + a_n v_n.$$

Explain why Γ and Λ are inverses of each other.

Proof.

$$\Lambda\Gamma(v) = \Lambda(\varphi_1(v), \cdots \varphi_n(v)) = \varphi_1(v)v_1 + \cdots + \varphi_n(v)v_n = v,$$

$$\Gamma\Lambda(a_1, \cdots, a_n) = \Gamma(a_1v_1 + \cdots + a_nv_n) = (\varphi_1(a_1v_1 + \cdots + a_nv_n), \cdots, \varphi_n(a_1v_1 + \cdots + a_nv_n)) = (a_1, \cdots, a_n).$$

□

Problem 8 (3F-10). Suppose m is a positive integer.

(a) Show that $1, x-5, \cdots, (x-5)^m$ is a basis of $\mathcal{P}_m(\mathbb{R})$.

(b) What is the dual basis of the basis in (a)?

Solution. (a) For any $p \in \mathcal{P}_m(\mathbb{R})$, we can write uniquely

$$p(x) = \sum_{k=0}^{m} \frac{p^{(k)}(5)}{k!}(x-5)^k,$$

which is not hard to verify.

(b) From the formula in (a), we can see that the dual basis of $1, x-5, \cdots, (x-5)^m$ is

$$\varphi_k(p) = \frac{p^{(k)}(5)}{k!}, \ 0 \le k \le m.$$

Problem 9 (3F-11). Suppose $v_1, \cdots, v_n$ is a basis of V and $\varphi_1, \cdots, \varphi_n$ is the corresponding dual basis of V'. Suppose $\psi \in V'$. Prove that

$$\psi = \psi(v_1)\varphi_1 + \cdots + \psi(v_n)\varphi_n.$$

Proof. For any $v \in V$, we can write $v = \varphi_1(v)v_1 + \cdots + \varphi_n(v)v_n$. Applying ψ to both sides, we get

$$\psi(v) = \psi(\varphi_1(v)v_1 + \cdots + \varphi_n(v)v_n) = \varphi_1(v)\psi(v_1) + \cdots + \varphi_n(v)\psi(v_n),$$

which is exactly we want to prove. □

Problem 10 (3F-15). Define $T : \mathcal{P}(\mathbf{R}) \to \mathcal{P}(\mathbf{R})$ by

$$(Tp)(x) = x^2 p(x) + p''(x)$$

for each $x \in \mathbf{R}$.

(a) Suppose $\varphi \in \mathcal{P}(\mathbf{R})'$ is defined by $\varphi(p) = p'(4)$. Describe the linear functional $T'(\varphi)$ on $\mathcal{P}(\mathbf{R})$.

(b) Suppose $\varphi \in \mathcal{P}(\mathbf{R})'$ is defined by $\varphi(p) = \int_0^1 p$. Evaluate $(T'(\varphi))(x^3)$.

Solution. (a) For any $p \in \mathcal{P}(\mathbb{R})$, we have

$$(T'(\varphi))(p) = \varphi(Tp) = (Tp)'(4) = (x^2 p(x) + p''(x))'(4) = 8p(4) + 16p'(4) + 5p'''(4).$$

(b) For any $p \in \mathcal{P}(\mathbb{R})$, we have

$$(T'(\varphi))(p) = \varphi(Tp) = \int_0^1 (Tp)(x)\,\mathrm{d}x = \int_0^1 (x^2 p(x) + p''(x))\,\mathrm{d}x = \int_0^1 x^2 p(x)\,\mathrm{d}x + p'(1) - p'(0).$$

Hence, we have

$$(T'(\varphi))(x^3) = \int_0^1 x^5\,\mathrm{d}x + 3 - 0 = \frac{1}{6} + 3 = \frac{19}{3}.$$

Problem 11 (3F-18). Suppose V and W are finite-dimensional. Prove that the map that takes $T \in \mathcal{L}(V, W)$ to $T' \in \mathcal{L}(W', V')$ is an isomorphism of $\mathcal{L}(V, W)$ onto $\mathcal{L}(W', V')$.

Proof. Denote

$$\Phi :\ \mathcal{L}(V, W) \to \mathcal{L}(W', V'),\ T \mapsto T'.$$

It suffices to show that Φ is injective and surjective.

First, we prove that $\ker \Phi = 0$. For $T \in \ker \Phi$, we have $0 = \Phi(T)(\varphi) = T'(\varphi) = \varphi T$ for arbitraty $\varphi \in W'$. Hence for some $v \in V$, we have $\varphi(Tv) = 0$ for any $\varphi \in W'$, which implies $Tv = 0$. Since $v \in V$ is arbitraty, we have $T = 0$. Thus, $\ker \Phi = 0$.

Next, we will show that Φ is surjective. Since V and W are finitee-dimensional, we have $V \simeq V'', W \simeq W''$ .In fact, we can treat $v \in V$ as a linear functional on V' , such that $v(\varphi) = \varphi(v)$, $\forall \varphi \in V'$. For the reason above, we will write down $V = V''$ and $W = W''$.

First, We will prove for given $T \in \mathcal{L}(V, W)$, we have

$$T = T''.$$

Notice that for any $\varphi \in W'$ and $v \in V$, by the associative property, we have

$$\varphi(T(v)) = (\varphi \circ T)(v) \implies (T(v))(\varphi) = v(\varphi \circ T) = v(T'(\varphi)).$$

(Remember that $\varphi \in W'$ and $Tv \in W = W''$; $\varphi \circ T \in V'$ and $v \in V = V''$.) Since

$$v(T'(\varphi)) = (v \circ T')(\varphi),$$

we have $(T(v))(\varphi) = (v \circ T')(\varphi)$ for any $\varphi \in W'$, hence $T(v) = v \circ T'$ for any $v \in V$. Since $T''(v) = v \circ T'$ for any $v \in V$, we have $T = T''$.

Next, we will show that for any $S \in \mathcal{L}(W', V')$, there exists $T \in \mathcal{L}(V, W)$ such that $S = \Phi(T) = T'$. Take $T = S'$, we have

$$T' = S'' = S \in \mathcal{L}(W', V').$$

$\square$

Problem 12 (3F-23). Suppose V is finite-dimensional and $\varphi_1, \cdots, \varphi_m \in V'$. Prove that the following three sets are equal to each other.

(a) $\operatorname{span}(\varphi_1, \cdots, \varphi_m)$;

(b) $((\ker \varphi_1) \cap \cdots \cap (\ker \varphi_m))^0$;

(c) $\{\varphi \in V' : (\ker \varphi_1) \cap \cdots \cap (\ker \varphi_m) \subset \ker \varphi\}$.

Proof. Denote the three sets by A, B, C respectively. We will show that $A = B = C$.

First, we will show that $A \subset B$. For any $\varphi \in A$, we can write $\varphi = \lambda_1 \varphi_1 + \cdots + \lambda_m \varphi_m$ for some $\lambda_1, \cdots, \lambda_m \in \mathbb{F}$. For any $v \in (\ker \varphi_1) \cap \cdots \cap (\ker \varphi_m)$, we have $\varphi_k(v) = 0$ for $1 \leq k \leq m$. Then

$$\varphi(v) = \lambda_1 \varphi_1(v) + \cdots + \lambda_m \varphi_m(v) = 0,$$

which implies $\varphi \in B$.

Next, by definition, $B = C$.

Finally, we will show that $C \subset A$. Pick a maximum linearly independent subset of $\{\varphi_1, \cdots, \varphi_m\}$, say $\{\varphi_1, \cdots, \varphi_k\}$. Then

$$\operatorname{span}\{\varphi_1, \cdots, \varphi_m\} = \operatorname{span}\{\varphi_1, \cdots, \varphi_k\}.$$

First, we will show that $\bigcap_{i=1}^k \ker \varphi_i \subset \ker \varphi_j$ for any $k+1 \leq j \leq m$. For any $v \in \bigcap_{i=1}^k \ker \varphi_i$, we have $\varphi_i(v) = 0$ for $1 \leq i \leq k$. Since $\{\varphi_1, \cdots, \varphi_k\}$ is a maximum linearly independent subset of $\{\varphi_1, \cdots, \varphi_m\}$, we can write $\varphi_j = \lambda_1 \varphi_1 + \cdots + \lambda_k \varphi_k$ for some $\lambda_1, \cdots, \lambda_k \in \mathbb{F}$. Then

$$\varphi_j(v) = \lambda_1 \varphi_1(v) + \cdots + \lambda_k \varphi_k(v) = 0,$$

which implies $v \in \ker \varphi_j$.

So, it suffices to prove

$$\left\{\varphi \in V' : \bigcap_{i=1}^k \ker \varphi_i \subset \ker \varphi\right\} \subset \operatorname{span}(\varphi_1, \cdots, \varphi_k).$$

We can extend $\{\varphi_1, \cdots, \varphi_k\}$ to a basis of V' , say $\{\varphi_1, \cdots, \varphi_k, \psi_{k+1}, \cdots, \psi_n\}$. Pick the dual basis, say $v_1, \cdots, v_n$. Notice that for any $k+1 \leq j \leq n$ and $1 \leq i \leq k$, we have $\varphi_i(v_j) = 0$, which indicates $v_j \in \bigcap_{i=1}^k \ker \varphi_i \subset \ker \varphi$.

For any $\varphi \in V'$ such that $\bigcap_{i=1}^k \ker \varphi_i \subset \ker \varphi$, we will prove

$$\varphi = \sum_{i=1}^k (\varphi(v_i))\varphi_i \in \operatorname{span}\{\varphi_1, \cdots, \varphi_k\}.$$

That's because, for any $v = \sum_{i=1}^n \lambda_i v_i$, we have

$$\begin{aligned}\varphi(v) &= \sum_{i=1}^{k} \lambda_i \varphi(v_i) + \sum_{i=k+1}^{n} \lambda_i \overbrace{\varphi(v_i)}^{\text{This term is zero}} \\ &= \sum_{i=1}^{k} \left(\sum_{j=1}^{n} \delta_{ij}\lambda_j \right) \varphi(v_i) = \sum_{i=1}^{k} \left(\sum_{j=1}^{n} \varphi_i(v_j)\lambda_j \right) \varphi(v_i) \\ &= \sum_{i=1}^{k} \varphi(v_i)\varphi_i(v).\end{aligned}$$

□

Problem 13 (3F-26). Suppose V is finite-dimensional and Ω is a subspace of V'. Prove that

$$\Omega = \{v \in V : \varphi(v) = 0 \text{ for every } \varphi \in \Omega\}^0.$$

Proof. First, we will show that $\Omega \subset \{v \in V : \varphi(v) = 0 \text{ for every } \varphi \in \Omega\}^0$, i.e. for any $\varphi \in \Omega$ and $v \in V$ such that $\psi(v) = 0$ for every $\psi \in \Omega$, we have $\varphi(v) = 0$. That's trivial.

Next, we will show that $\{v \in V : \varphi(v) = 0 \text{ for every } \varphi \in \Omega\}^0 \subset \Omega$. □